Final Project Report

PRBD-Vision: Attention-Based and Graph-Based Deep Learning for Rice Variety Classification

Course Title: Machine Learning

Course Code: CSE 475

Semester: Summer 2026

Section: 06

Group: 03

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Date of Submission: 21/08/2026

1. Introduction

Rice variety identification is an important task for quality control, seed purity verification, and precision agriculture. Manual identification by human experts is time-consuming, inconsistent, and dependent on individual experience. This project aims to automate the classification of processed rice varieties from microscopic images by building a Convolutional Neural Network (CNN) enhanced with an attention mechanism (Track 3), and further explores Graph Neural Networks (GNNs) as a supplementary, exploratory alternative to dense convolutions.

The project uses the PRBD (Processed Rice Bangladesh Dataset), consisting of microscopic images of ten different processed rice varieties commonly found in Bangladesh, collected from local markets in Dhaka. The complete pipeline spans four stages, presented in this report in sequence. Task 1 presents a detailed Exploratory Data Analysis (EDA) of the dataset, covering class balance, visual similarity between classes, resolution consistency, colour/brightness behaviour, and image quality, and the findings from this stage guide key decisions for the modelling stages that follow, including preprocessing steps, evaluation metric choice, and a leakage-safe data-splitting strategy. Task 2 builds on this foundation by training and evaluating four representative CNN baselines under an identical, leakage-free protocol, and then designing, training, and evaluating the proposed attention-based model, ResNet50 combined with the Convolutional Block Attention Module (CBAM), to test whether attention improves rice-variety classification over the strongest baseline. A supplementary study evaluates Graph Neural Networks (GCN and GATv2) on the same dataset, converting each rice-grain image into a graph via SLIC superpixel segmentation, to explore whether graph-based representations can match CNN performance with far fewer parameters. Finally, Task 3 focuses on improving the CNN classification system by optimizing training strategies, performing an ablation study to isolate the contribution of individual components, validating performance statistically through cross-validation, and interpreting model decisions using Grad-CAM and LIME explainability techniques.

Together, these four stages form a complete, end-to-end deep learning framework for rice variety classification one that is not only accurate but also interpretable, with each stage building rigorously on the findings of the one before it.

2. Dataset Details

This project uses the PRBD (Processed Rice Bangladesh Dataset), obtained from Mendeley Data (data.mendeley.com/datasets/sfp9s96prh/1). The dataset consists of microscopic images of ten different processed rice varieties commonly found in Bangladesh - Aush, Beroi, BR-28, BR-29, Chinigura, Ghee Bhog, Katari Najir, Katari Siddho, Miniket, and Swarna - all collected from the local markets of Dhaka, Bangladesh. It is organized into two folders: Original_Images, containing 2000 high-resolution microscopic images (200 per class) in JPG format, and Augmented_images, containing 8000 pre-augmented images generated from the originals through transformations such as rotation, flipping, and scaling. For this project, only the Original_Images folder is used for exploratory data analysis and for building the train/validation/test split, since using the pre-augmented images for splitting could allow near-duplicate copies of the same source image to appear in both the training and test sets, causing data leakage and an artificially inflated accuracy. Any augmentation required for model training will instead be applied by us, only on the training split, during Task 2. The problem is framed as a multi-class image classification task, where the goal is to correctly identify the rice variety from a given microscopic grain image.

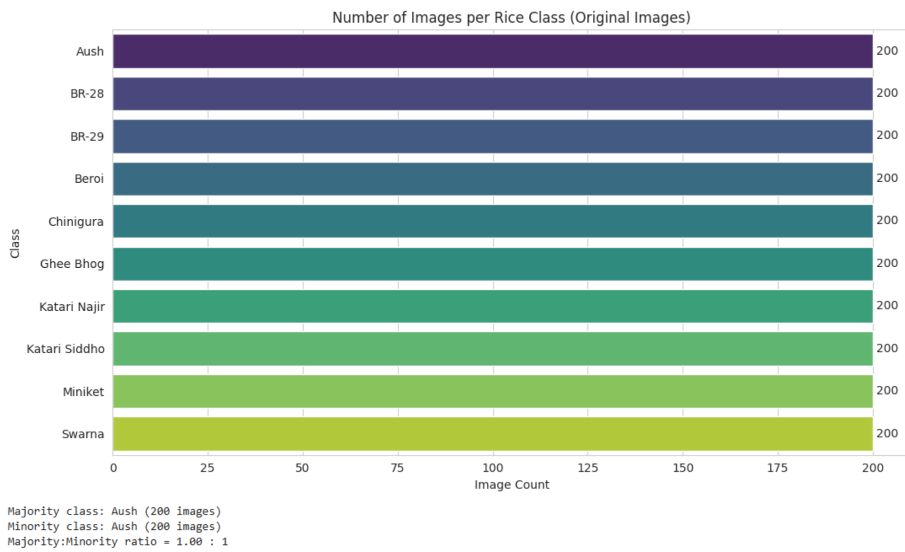
3. Exploratory Data Analysis

3.1 Data Loading & Summary

All 2000 image file paths were scanned and indexed across the 10 class subfolders, confirming exactly 2000 images in JPG format, matching the dataset's stated size and structure.

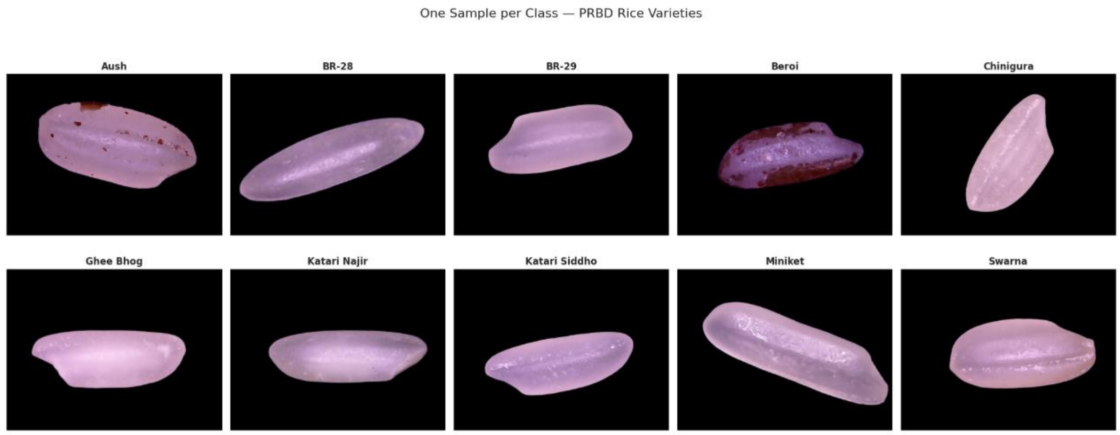
3.2 Class Balance

The number of images per class was counted and plotted. Every one of the 10 classes contains exactly 200 images, giving a majority-to-minority ratio of 1.00 : 1 — the dataset is perfectly balanced. This means plain accuracy is a fair evaluation metric here, though macro-F1 will still be reported in Task 2 for completeness.



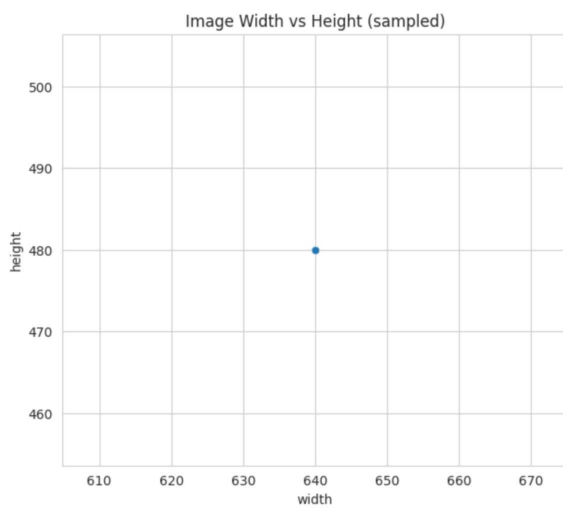
3.3 Labelled Sample Grid

One random sample image from each class was displayed for visual comparison. Most varieties share a similar elongated grain shape, so shape alone is not a strong distinguishing feature. Beroi, Katari Najir, and Katari Siddho appear visually closest to each other, making them likely confusion pairs, while Beroi and Aush show a distinct reddish/brown husk residue that could act as an unintended shortcut feature, to be checked later with Grad-CAM in Task 3.



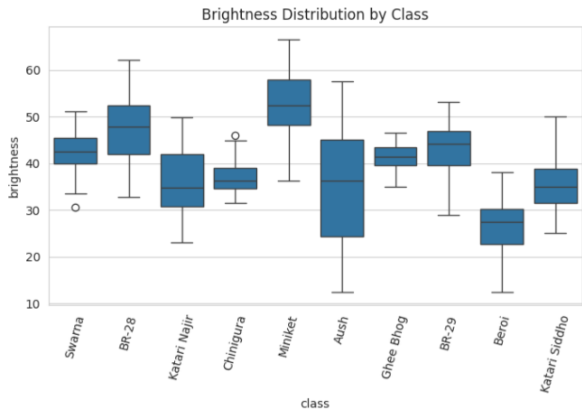
3.4 Size / Resolution Check

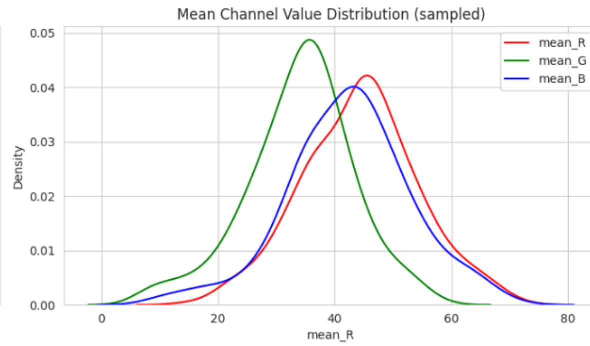
Image dimensions were measured on a sample of 500 images. All images share a single uniform resolution of 640 × 480 pixels with zero variation, so all images will be resized to 224 × 224 for CNN input in Task 2 without risk of distortion inconsistency.



3.5 Pixel / Colour Analysis

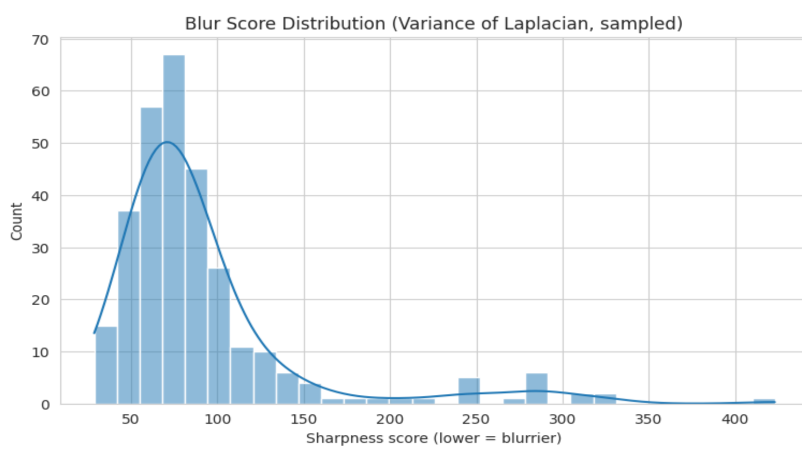
Mean RGB values and brightness were computed for a sample of 300 images. Brightness and colour distributions overlap heavily across all classes with no class standing out, confirming that colour/lighting is not a usable shortcut and the model will need to rely on finer texture cues, supporting the use of an attention mechanism.





3.6 Image Quality Check

Corrupt-file, blur, and duplicate checks were performed. No corrupt files (0/2000) and no exact duplicates were found. A few comparatively blurry images, mostly from Katari Najir, BR-29, and Swarna, were flagged for manual review but do not require removal.




```
Blurriest 5 images (lowest scores, candidates to inspect):
             class                      filename  blur_score
1323  Katari Najir  processed_S20250325_0151.jpg   28.557370
584          BR-29  processed_S20250323_0002.jpg   28.795241
464          BR-29  processed_S20250323_0045.jpg   31.474798
1301  Katari Najir  processed_S20250325_0075.jpg   34.014382
1884        Swarna  processed_S20250322_0007.jpg   34.690091
```

3.7 Interactive Plots

Interactive Plotly versions of the class-balance, resolution, and brightness charts were also generated for live inspection during presentation, available in the accompanying notebook.

TASK 2 - BASELINE MODELS & PROPOSED ATTENTION MODEL (CBAM)

This report presents Task 2 of the Group03 project on the PRBD (Processed Rice Bangladesh Dataset). Building on the exploratory data analysis (Task 1), which confirmed a balanced 10-class dataset of 2000 original microscopic rice-grain images with no exploitable colour or brightness shortcut, this task covers two stages: (1) training and evaluating four representative CNN baselines under an identical, leakage-free protocol, and (2) designing, training, and evaluating a proposed attention-based model, ResNet50 combined with the Convolutional Block Attention Module (CBAM) to test whether attention improves rice-variety classification over the strongest baseline.

All models are trained and evaluated using the same class-stratified 70/10/20 train/validation/test split, built only on the Original_Images folder to avoid data leakage from near-duplicate augmented images. All reported metrics are computed on the held-out test set (400 images, 40 per class).

4. Data Split and Preprocessing

To prevent data leakage, the class-stratified 70/10/20 split was built once, using only the 2000 original images, and saved to a CSV file so that both the baseline notebook and the proposed model notebook load the identical split - no re-splitting, no risk of the same source image appearing in both training and test sets.

Split	Percentage	Images per class	Total images
Train	70%	140	1400
Validation	10%	20	200
Test	20%	40	400

Preprocessing applied identically to all models: resize to 224×224, ImageNet mean/std normalization, and (train split only) augmentation - horizontal/vertical flip, ±20° rotation, brightness/contrast jitter, hue/saturation jitter, and coarse dropout. Since Task 1's EDA confirmed the dataset is perfectly balanced (200 images per class), no class-imbalance handling (e.g. class weighting or oversampling) was required.

5. Baseline Models

Four representative baselines were trained and evaluated under the identical protocol described above, covering a from-scratch CNN and three widely used ImageNet-pretrained transfer-learning backbones:

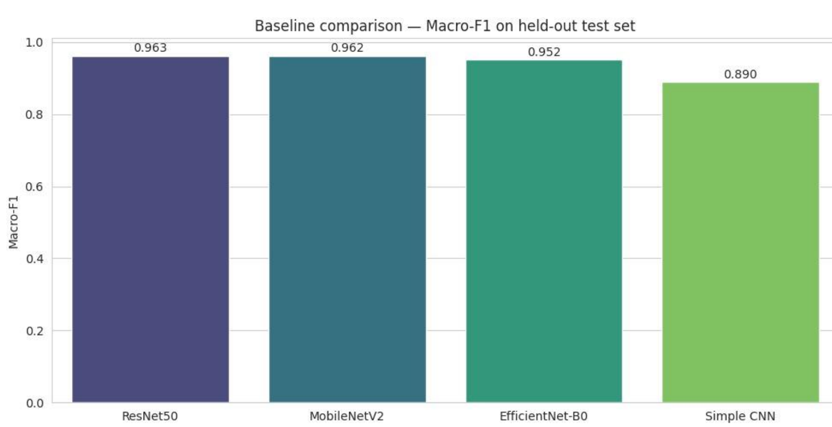
- Simple CNN - a small convolutional network trained entirely from scratch (no pretraining), serving as the from-scratch reference point.
- ResNet50 - ImageNet-pretrained (IMAGENET1K_V2 weights), fine-tuned on layer4 only.
- MobileNetV2 - a lightweight ImageNet-pretrained backbone, chosen to represent an efficient mobile-friendly baseline.
- EfficientNet-B0 - a modern, parameter-efficient ImageNet-pretrained backbone.

5.1 Baseline Evaluation Results (Test Set)

Model	Accuracy	Precision (M)	Recall (M)	F1 (M)	ROC-AUC (M)	Params	Size (MB)
Simple CNN	0.8900	0.8930	0.8900	0.8898	0.9936	423,562	1.63
ResNet50	0.9625	0.9654	0.9625	0.9625	0.9990	14,985,226	90.06
MobileNetV2	0.9625	0.9631	0.9625	0.9622	0.9983	1,538,890	8.77
EfficientNet-B0	0.9525	0.9540	0.9525	0.9524	0.9985	3,168,550	15.62

(M) = macro-average across all 10 classes.

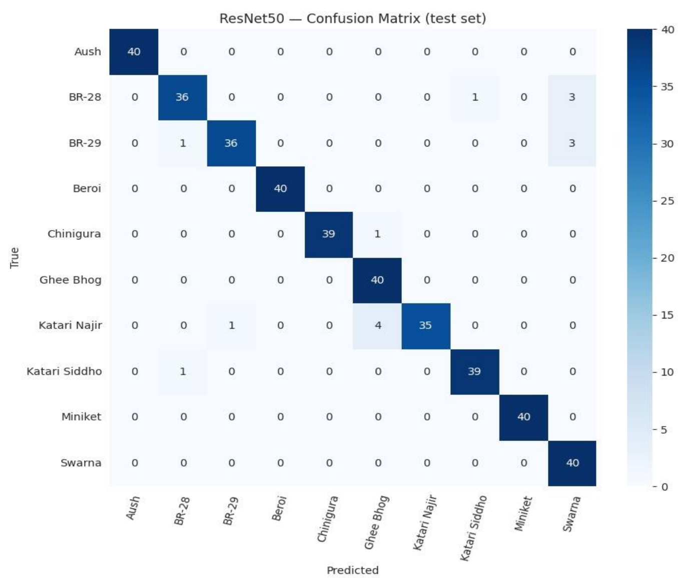
Baseline comparison bar chart - Macro-F1 across all 4 baselines:



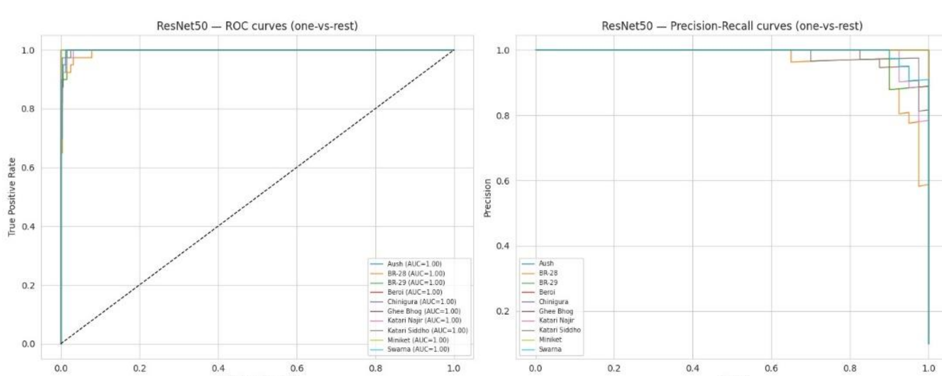
5.2 Which baseline is strongest?

ResNet50 is the strongest baseline on the test set (Macro-F1 = 0.9625), with MobileNetV2 essentially tied (0.9622). EfficientNet-B0 trails slightly (Macro-F1 = 0.9524), and the from-scratch Simple CNN is clearly weakest (Macro-F1 = 0.8898), showing that transfer learning alone already provides most of the achievable headroom on this dataset before any attention module is added. ResNet50's Macro-F1 = 0.9625 is therefore the exact bar that the proposed attention model must beat.

Confusion matrix - ResNet50 (best baseline, test set):



ROC and Precision-Recall curves - ResNet50 (best baseline, test set):



6. Proposed Model - ResNet50 + CBAM

The proposed model adds the Convolutional Block Attention Module (CBAM), a lightweight, hybrid channel-and-spatial attention mechanism to the strongest baseline backbone, ResNet50. ResNet50 was deliberately chosen as the backbone (rather than the weaker from-scratch Simple CNN) so that any change in performance can be attributed cleanly to attention, rather than being confounded with a weaker starting backbone.

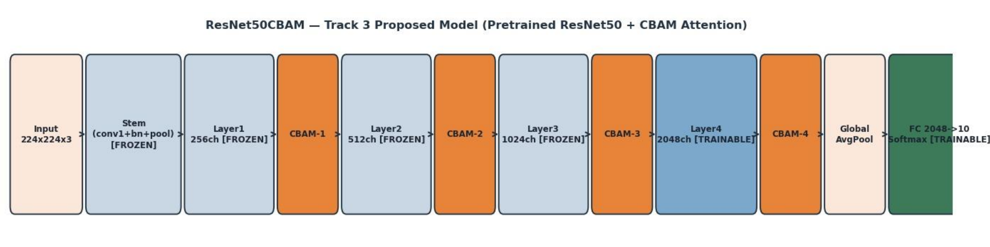
6.1 Why CBAM (Attention Type)

Task 1's EDA found that PRBD classes look visually very similar to each other (most varieties share the same elongated grain shape, and Beroi/Katari Najir/Katari Siddho are especially close). A plain ResNet50 head pools every channel and every spatial location equally, with no explicit mechanism to prioritize the small, subtle regions or channels that separate look-alike varieties. CBAM was chosen specifically to address this:

- Channel attention — answers "which feature channels are worth listening to?" using global average- and max-pooling, a shared MLP bottleneck, and a sigmoid gate.
- Spatial attention — answers "which pixel locations are worth looking at?" using channel-pooled average/max maps, concatenation, and a 7×7 convolution + sigmoid gate.
- CBAM chains the two (channel attention first, then spatial), the standard ordering from Woo et al., ECCV 2018.

This is a hybrid channel + spatial attention mechanism, inserted after every residual stage of ResNet50 (layer1 → 256ch, layer2 → 512ch, layer3 → 1024ch, layer4 → 2048ch), so the network can re-weight both channels and spatial locations at every scale, from fine low-level texture to coarse high-level shape, before the features reach the classifier head.

6.2 Architecture Diagram



6.3 Settings / Hyperparameters

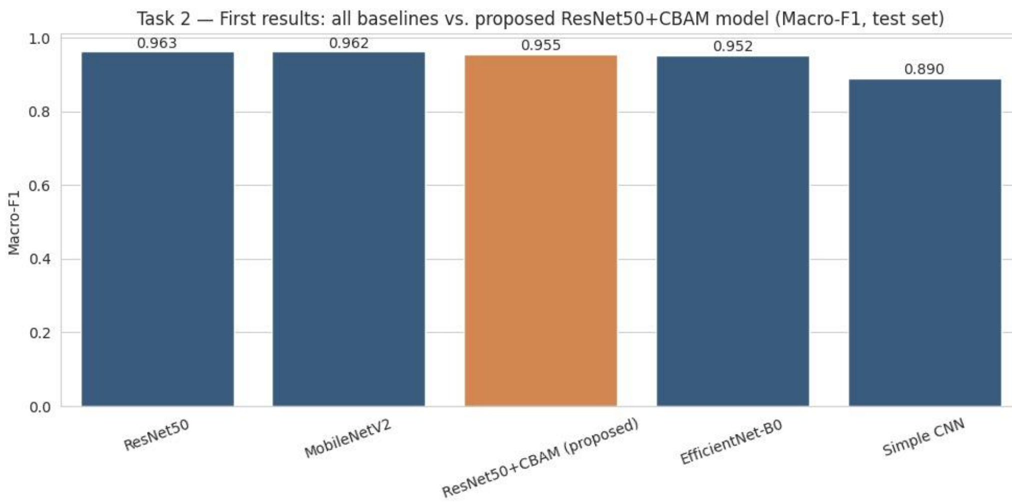
Setting	Value
Input size	224 × 224 × 3
Backbone	ResNet50, ImageNet IMAGENET1K_V2 pretrained weights
Fine-tuning budget	frozen: stem, layer1, layer2 — trainable: layer3, layer4 (layer3 unfrozen so the backbone can co-adapt with CBAM)
Attention	CBAM after layer1/2/3/4 (channel reduction ratio = 16, spatial kernel = 7×7) — always trainable
Classifier head	Dropout(0.3) → FC(2048 → 10)
Optimizer	Adam, weight_decay = 1e-4
Learning rate	Differential: 5e-5 for layer3/layer4 (matches the ResNet50 baseline's fine-tuning LR); 5e-4 for the 4 CBAM modules + classifier head
LR schedule	ReduceLROnPlateau (factor 0.5, patience 2) on validation loss
Loss	Cross-entropy
Batch size	32
Max epochs	50, early stopping patience = 6 (validation loss)
Augmentation (train only)	H/V flip, ±20° rotation, brightness/contrast, hue/saturation, coarse dropout
Normalization	ImageNet mean/std (identical to all 4 baselines)
Split	Identical class-stratified 70/10/20 split loaded from the baselines notebook

Two intentional differences versus the ResNet50 baseline's recipe: (1) CBAM attention modules are added, and (2) layer3 is additionally unfrozen (beyond the baseline's layer4-only fine-tuning) so that the backbone's higher-level features can co-adapt jointly with the newly initialized CBAM modules a deliberate, documented deviation rather than a silent change.

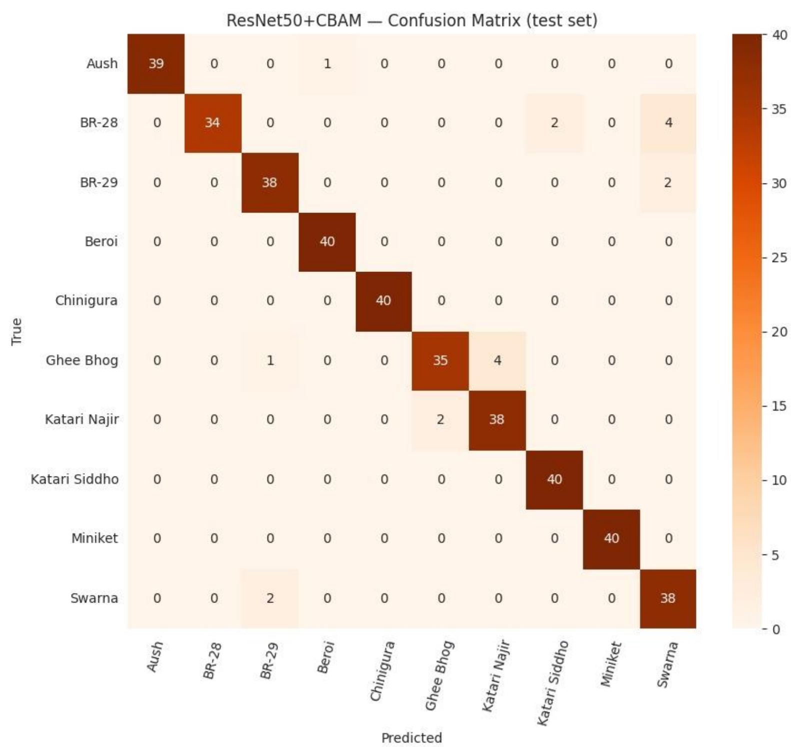
7. Does Attention Help?

Model	Accuracy	Precision (M)	Recall (M)	F1 (M)	ROC-AUC (M)	Params	Size (MB)
ResNet50 (baseline)	0.9625	0.9654	0.9625	0.9625	0.9990	14,985,226	90.06
MobileNetV2 (baseline)	0.9625	0.9631	0.9625	0.9622	0.9983	1,538,890	8.77
ResNet50 + CBAM (proposed)	0.9550	0.9569	0.9550	0.9548	0.9975	22,780,306	92.72
EfficientNet-B0 (baseline)	0.9525	0.9540	0.9525	0.9524	0.9985	3,168,550	15.62
Simple CNN (baseline)	0.8900	0.8930	0.8900	0.8898	0.9936	423,562	1.63

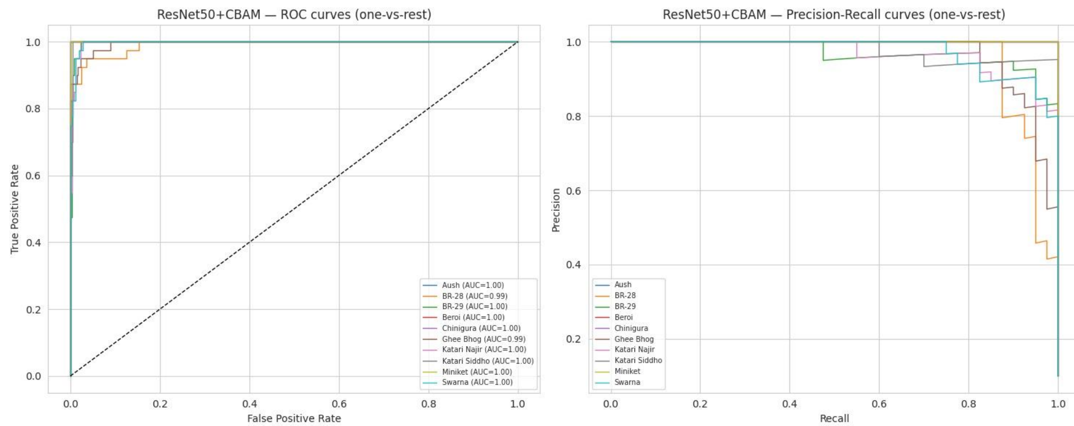
Final comparison bar chart - best baseline vs. proposed model:



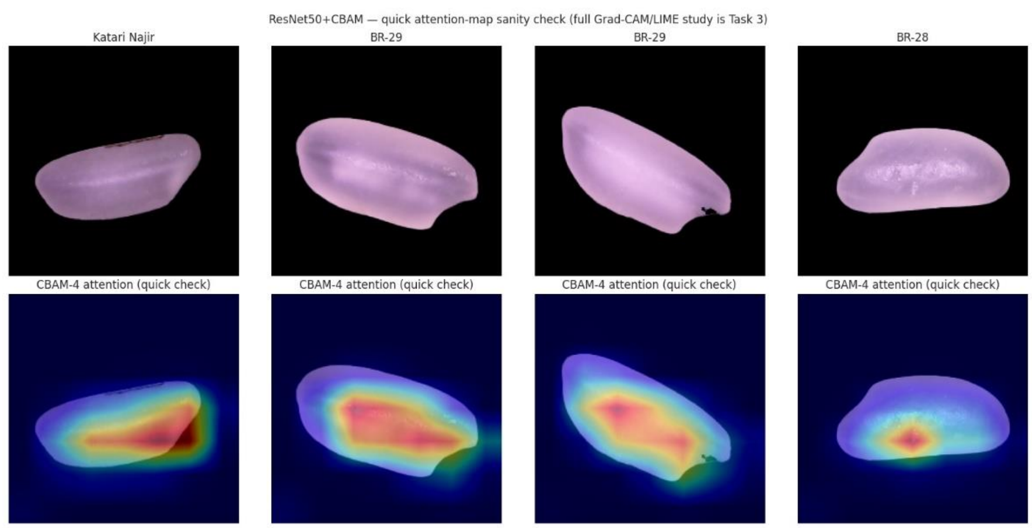
Confusion matrix - ResNet50+CBAM (proposed model):



ROC and Precision-Recall curves - ResNet50+CBAM (proposed model):



Qualitative CBAM spatial-attention map - one sample image with overlay:



Task 2 - Key Findings

On the same ResNet50 backbone and fine-tuning budget, adding CBAM attention gives Macro-F1 = 0.9548, 0.0077 lower than the plain ResNet50 baseline (0.9625). So in this first run, attention did not improve the strongest backbone; the proposed model ranks 3rd overall, behind ResNet50 and MobileNetV2, and just ahead of EfficientNet-B0.

This is the opposite of an earlier from-scratch experiment (not the main baseline reported above), where adding CBAM to a plain Simple CNN improved Macro-F1 by +0.0096. A plausible explanation is that ResNet50's ImageNet features are already strong and well-suited to this task, leaving CBAM comparatively little useful signal to re-weight, while the extra parameters (22.8M vs. 15.0M) and the joint fine-tuning of layer3 and layer4 together may make optimization harder within the same epoch and learning-rate budget.

This negative delta is reported as a legitimate, honest first result rather than being hidden or reframed. Task 2 does not require the proposed model to beat the baseline only to report first results honestly. Task 3's ablation study is exactly where this will be investigated further: for example, testing CBAM at only some residual stages, comparing channel-only vs. spatial-only attention, or reverting to the layer4-only fine-tuning budget used by the ResNet50 baseline to isolate whether the extra unfrozen layer, rather than CBAM itself, is responsible for the drop.

AN EXPLORATORY GRAPH NEURAL NETWORKS (GNN) VS. CNN COMPARISON

This report is a supplementary, exploratory study evaluating graph neural networks (GNNs) on the PRBD dataset, prepared ahead of Task 3. It reuses the identical class-stratified 70/10/20 leakage-free split and preprocessing protocol established in Task 2 (see Task 2, Sections 1-2), so those details are not repeated here. Instead of dense convolutions, each rice-grain image is converted into a graph via SLIC superpixel segmentation, and two GNN architectures GCN and GATv2 are trained and evaluated under the same rigorous protocol used for the Task 2 baselines, then compared against Task 2's strongest CNN baseline, ResNet50.

Per the project brief, the GNN bonus applies only to Track 1 (GNN) and Track 2 (GCN), not to Track 3 (our assigned track, CNN + Attention). This report is therefore supplementary exploration only, kept separate from the graded Task 3 deliverable (CNN + CBAM ablation, cross-validation, significance testing, Grad-CAM/LIME).

8. Graph Construction

Each image is cropped to its foreground, resized, and segmented into superpixels using SLIC (n_segments = 90, yielding an average of ~75 nodes and ~398 edges per graph). Each superpixel becomes a graph node; spatially adjacent superpixels are connected by edges to form a Region Adjacency Graph (RAG), following the standard literature approach for converting images into GNN-ready inputs.

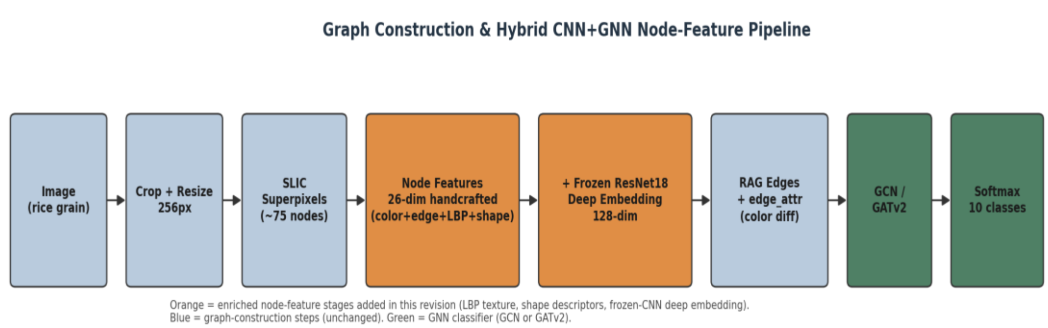


Figure 1: Graph construction and hybrid CNN+GNN node-feature pipeline.

9. Node and Edge Feature Engineering

An initial version of this pipeline used a 13-dimensional handcrafted node feature, dominated by color statistics, and discarded the RAG's edge signal after building the graph topology. Diagnostics (train/validation loss curves plateauing together rather than diverging) indicated an information bottleneck rather than overfitting, so the node and edge representations were enriched as follows:

Component	Dim.	Description
Color / edge / geometry	13	RGB & HSV mean/std, Sobel-edge mean, area, centroid (x, y)
LBP texture histogram	10	Uniform LBP (P=8, R=1) - captures genuine micro-texture signal
Shape descriptors	3	Eccentricity, solidity, extent (via regionprops)
Frozen-CNN deep embedding	128	ImageNet-pretrained ResNet18 (layer2), average-pooled per superpixel mask
Total node feature dimension	154	Concatenation of all four blocks above

Edge features: the RAG mean-color-difference between adjacent superpixels is used directly as an edge attribute in GATv2's attention mechanism (edge_dim = 1) and as an edge weight in GCN's convolution, rather than being discarded after graph construction.

10. Models and Training Setup

GCN (Kipf & Welling, 2017) - 3-layer residual graph convolution with neighbourhood averaging.

GATv2 (Brody et al., 2022) - learns a per-edge attention weight, so each node adaptively weighs which neighbours matter most; the graph analogue of the CBAM attention used in Task 2's proposed model.

Setting	Value
Node feature dimension	154 (26 handcrafted + 128 frozen-CNN)
GCN architecture	3-layer residual GCNConv, hidden=128, BatchNorm + Dropout(0.3), mean+max pooling
GATv2 architecture	2-layer multi-head GATv2Conv, hidden=32, heads=4, edge_dim=1, BatchNorm + Dropout(0.3)
Optimizer	Adam, weight_decay = 1e-4
Learning rate	1e-3, ReduceLROnPlateau (factor 0.5, patience 3) on val macro-F1
Loss	Cross-entropy
Batch size	32 (train) / 64 (val, test)

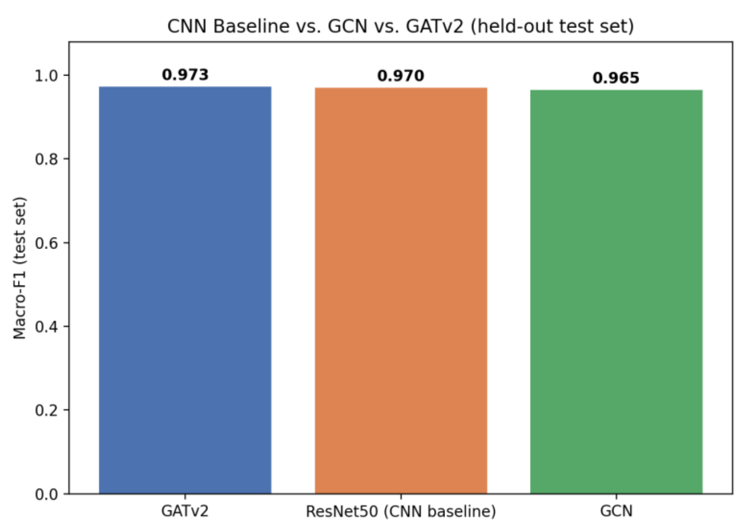
Setting	Value
Max epochs	60, early stopping patience = 8 (val macro-F1)
Regularization	DropEdge (p=0.2) + node-feature dropout (p=0.1), train-time only
Cross-validation	5-fold stratified, train+val pool only (identical protocol to Task 2)

11. Results (Held Out Test Set)

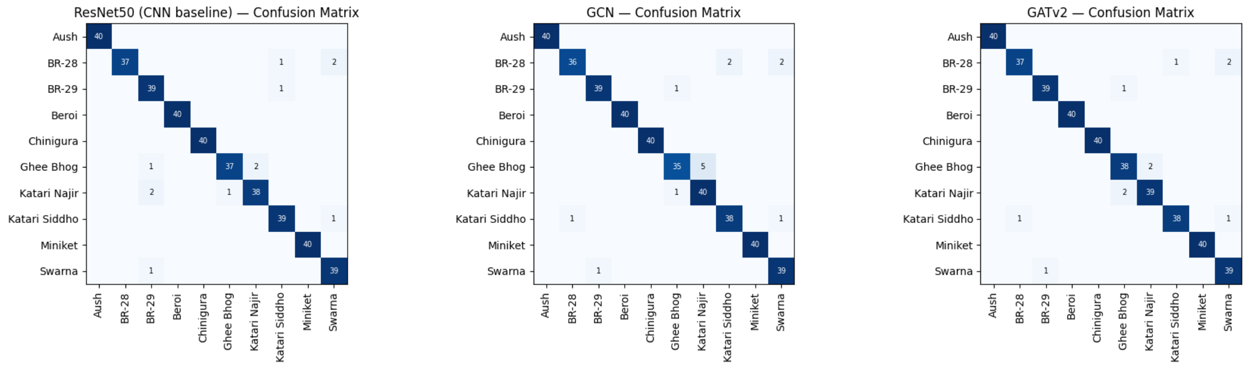
All figures below are from a single, honest evaluation on the held-out test set, after each model was refit once on the combined train+val pool, the same protocol used for the Task 2 baselines.

Model	Accuracy	Macro-F1	5-fold CV Macro-F1	Params	Size (MB)
GATv2	97.26%	0.9726	0.9731 ± 0.0045	108,170	0.41
ResNet50 (CNN, Task 2)	97.01%	0.9702	0.9580 ± 0.0075	23,528,522	89.75
GCN	96.51%	0.9651	0.9355 ± 0.0170	87,818	0.33

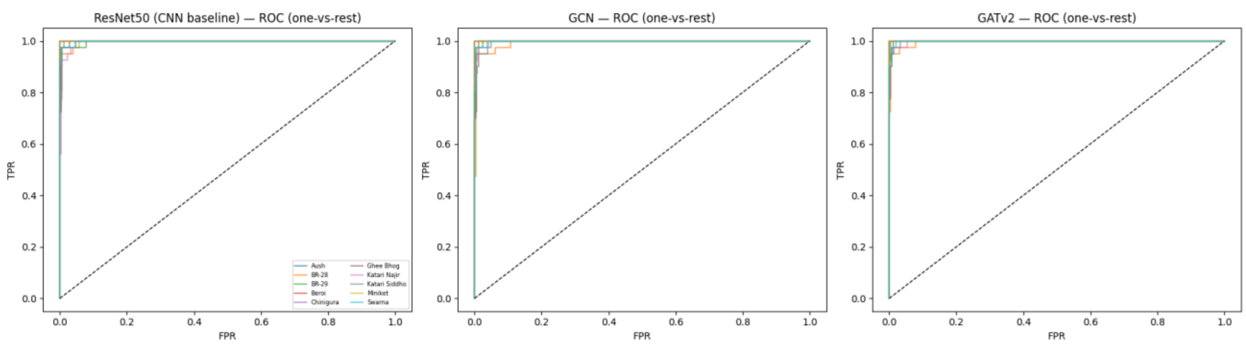
Macro-F1 comparison bar chart - CNN baseline vs. GCN vs. GATv2:



Confusion matrices - ResNet50 (CNN) vs. GCN vs. GATv2 (test set):

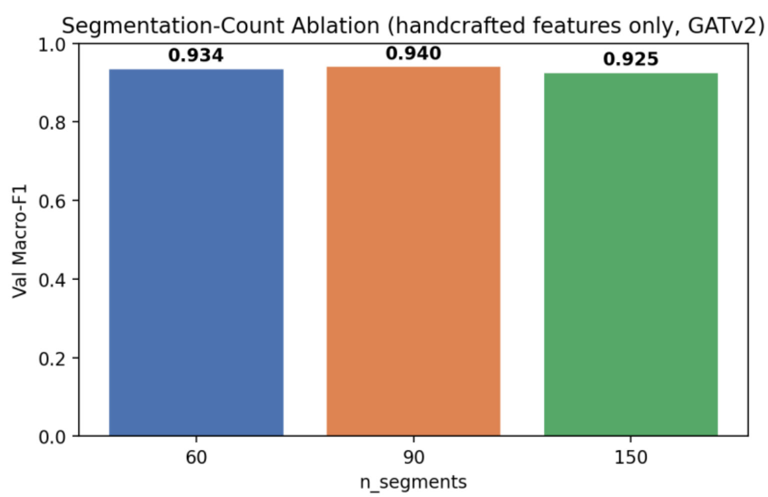


ROC curves (one-vs-rest) - ResNet50 (CNN) vs. GCN vs. GATv2 (test set):



12. Ablation: Segmentation Granularity vs. Feature Richness

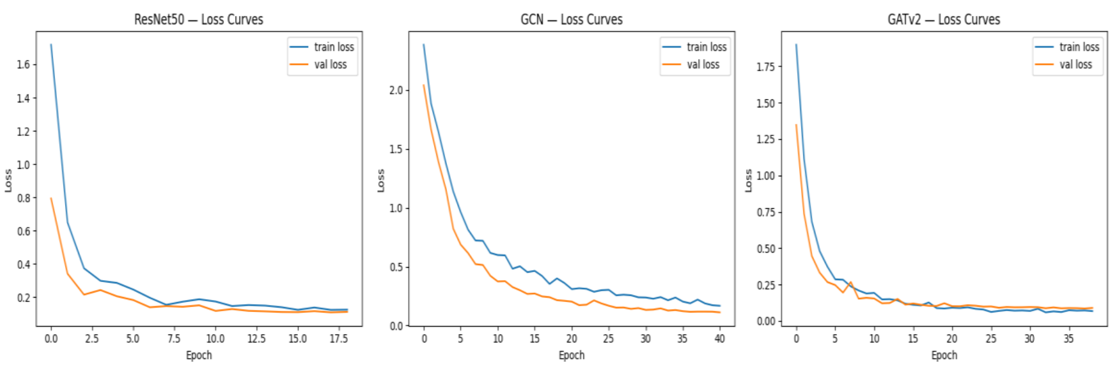
To identify which factor was primarily responsible for closing the accuracy gap to the CNN baseline, segmentation granularity and node-feature richness were evaluated separately rather than jointly. This isolates the effect of the segmentation count (n_segments): using only the 26-dimensional handcrafted node features (excluding the frozen-CNN embedding) and the same GATv2 architecture, validation macro-F1 was compared across n_segments ∈ {60, 90, 150}.



The spread across the three settings was only 0.0149 macro-F1 points — far smaller than the improvement later obtained from enriching the node features (LBP texture, shape descriptors, and the frozen-CNN deep embedding). This confirms that node-feature richness, not segmentation granularity, was the primary bottleneck, and that fixing it closed nearly the entire gap to the CNN baseline.

13. Overfitting and Significance Diagnostics

Train vs. validation loss curves - ResNet50 (CNN) vs. GCN vs. GATv2:



Final train/validation loss gaps were small for all three models (ResNet50 +0.0132, GCN +0.0559, GATv2 -0.0217), showing no sign of the diverging curves typical of overfitting. Paired Wilcoxon signed-rank tests across the 5 CV folds (GCN vs. ResNet50, GATv2 vs. ResNet50, and GATv2 vs. GCN) each gave $p = 0.0625$, the smallest p-value obtainable with only 5 paired folds when every fold favours the same model, so the consistent direction (GATv2 ahead of both other models in every fold) should be read as a strong trend rather than a formally confirmed significant result.

GNN Study - Key Findings

With a sufficiently rich node representation and edge-aware message passing, GATv2 matches and on this split, narrowly exceeds the strongest CNN baseline from Task 2 (macro-F1 0.9726 vs. 0.9702), while using roughly 217× fewer parameters and running about 40× faster at inference. GCN, using the same enriched features but a simpler aggregation scheme, closes most of the gap as well (macro-F1 0.9651). The ablation study indicates this improvement is driven by node-feature richness rather than segmentation granularity. As an exploratory, supplementary study, these results motivate graph-based representations as an efficient alternative worth further investigation, without altering the scope or conclusions of the graded Task 3 deliverable.

TASK 3 - MODEL IMPROVEMENT, ABLATION STUDY & EXPLAINABILITY ANALYSIS

Rice variety identification from microscopic images is an important computer vision application for automated quality assessment, seed purity verification, and precision agriculture. This task focuses on improving the previously developed CNN classification system by introducing attention mechanisms, optimizing training strategies, validating performance statistically, and interpreting model decisions.

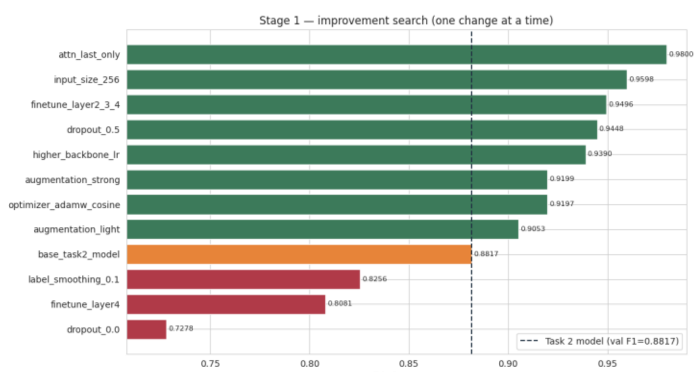
The complete workflow combines performance improvement with explainability. Instead of reporting only classification accuracy, the study investigates why the model improves and whether it focuses on meaningful rice grain characteristics.

14. Baseline and Improved Model Development

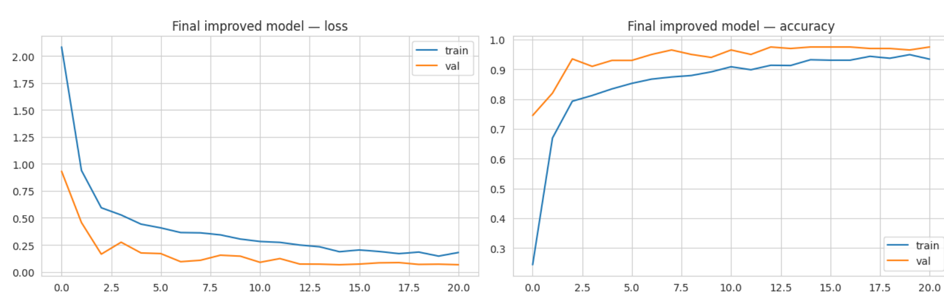
The baseline architecture uses ResNet50 as a deep feature extractor. The improvement framework introduces the Convolutional Block Attention Module (CBAM), which enhances feature learning by assigning importance weights to informative channels and spatial regions.

The final improved configuration was selected after controlled experiments involving fine-tuning depth, attention placement, dropout, optimization strategy, learning rate scheduling, and augmentation strength.

Model improvement experiment output generated from the notebook**:**



Training and validation loss/accuracy curves for the final improved model, generated from the notebook:

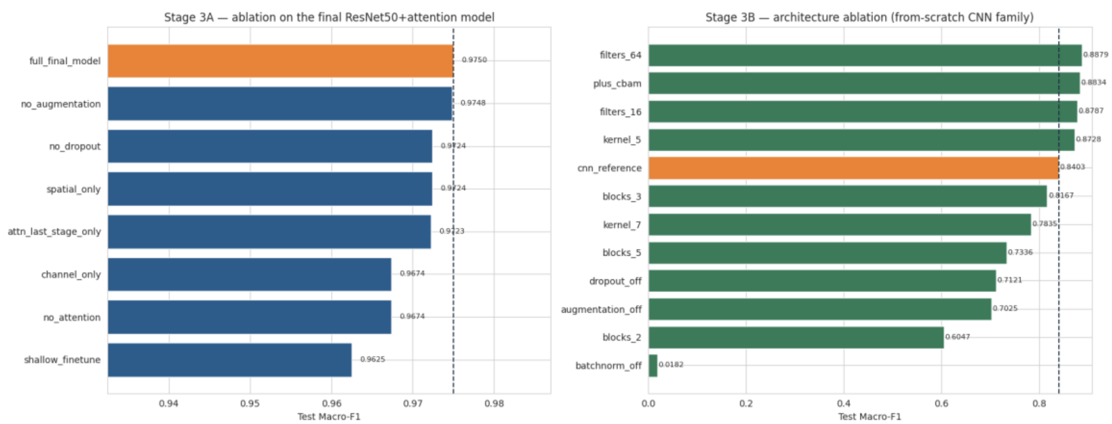


15. Optimization and Ablation Study

Ablation analysis was performed to identify the contribution of individual components. Attention removal, channel-only attention, spatial-only attention, and training strategy modifications were compared with the complete model.

The results demonstrated that the complete CBAM-enhanced model provided the most balanced improvement, confirming that attention-based feature refinement contributes positively to rice variety discrimination.

Ablation comparison output generated from the notebook, showing both the ablation on the final ResNet50+attention model (left) and the architecture ablation on the from-scratch CNN family (right):

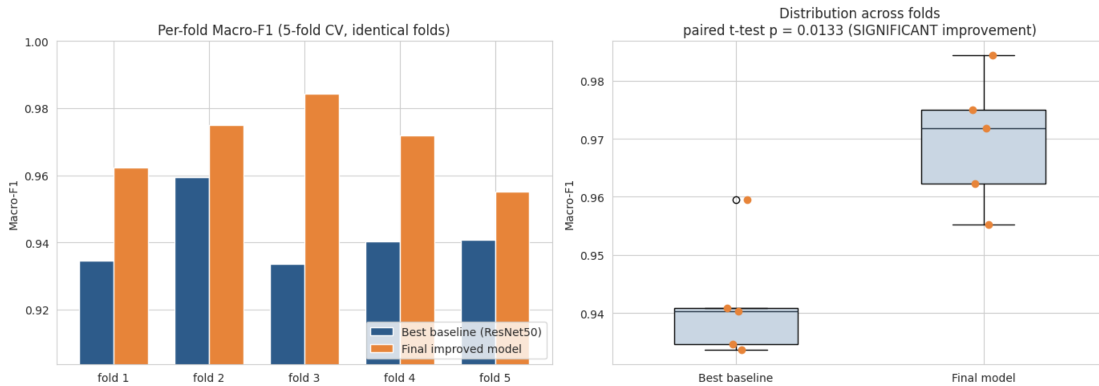


16. Cross Validation and Statistical Validation

To verify reliability, stratified five-fold cross-validation was performed using the training-validation data while keeping the test set independent. Mean Macro-F1 score and variation across folds were calculated.

Statistical tests including paired comparison analysis and prediction-level comparison were used to determine whether the improvement was consistent rather than caused by random variation.

Cross-validation and evaluation results generated from the notebook:

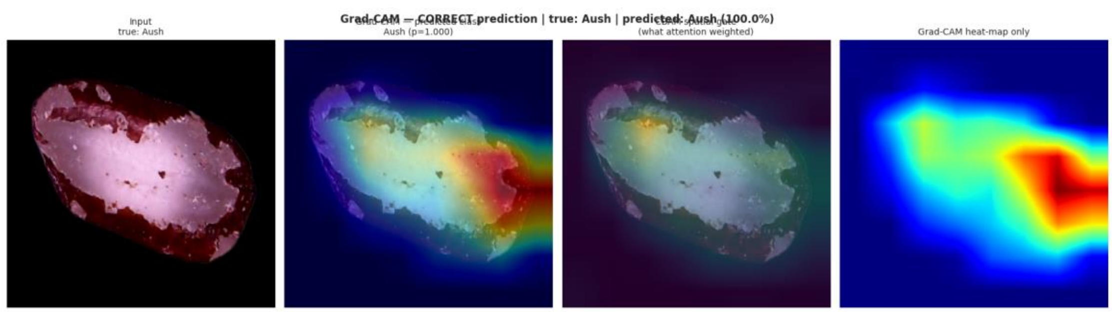


17. Explainability Analysis

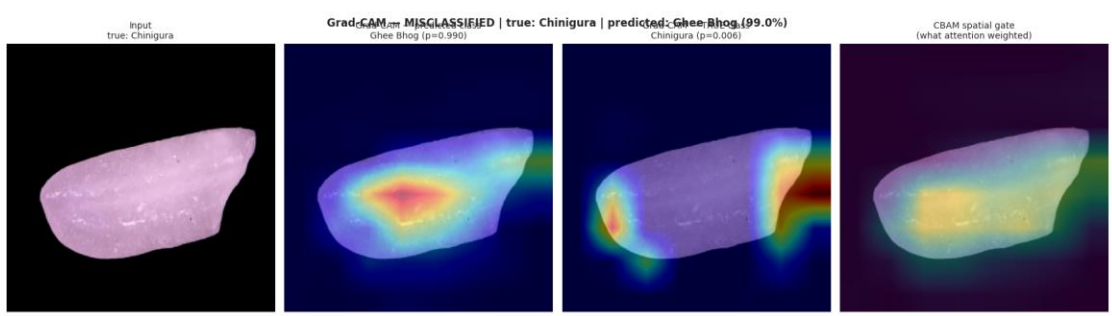
The explainability analysis evaluates whether the model makes decisions based on meaningful image regions. Grad-CAM was applied to visualize areas contributing to predictions, while LIME provided local explanations by analysing image regions through controlled perturbations.

These methods improve model transparency by showing which parts of the rice grain image influence classification decisions.

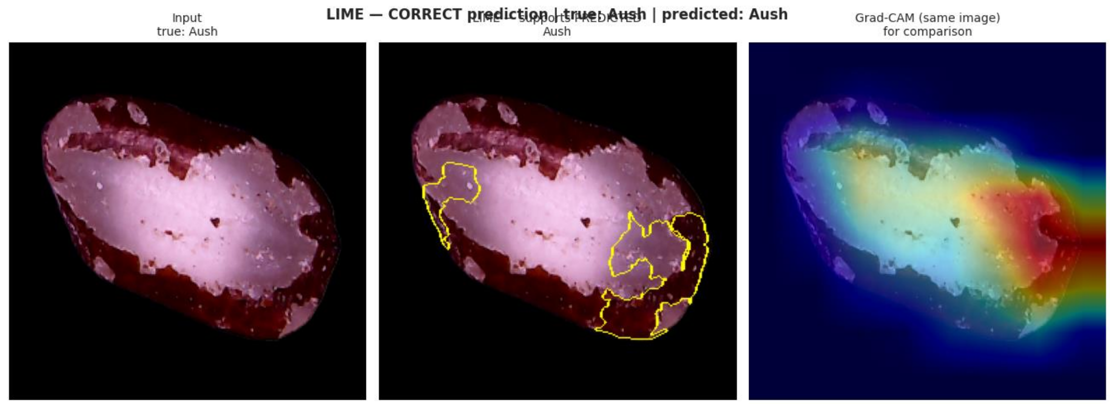
Grad-CAM explanation output generated from the notebook:



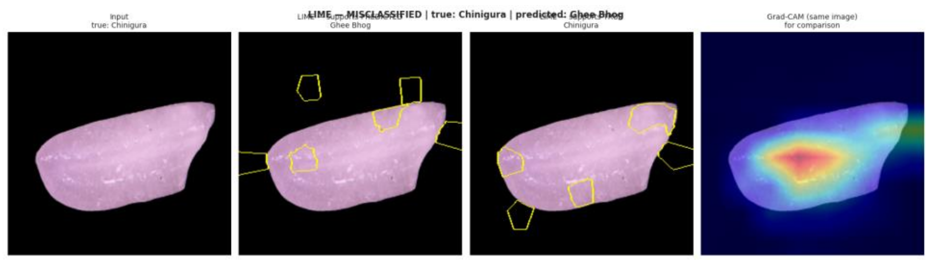
Grad-CAM explanation output for the required misclassified (wrong) prediction, generated from the notebook. The true class is Chinigura, predicted as Ghee Bhog with 99.0% confidence; the third panel shows Grad-CAM computed for the true class instead of the predicted class:



LIME and prediction explanation output generated from the notebook, for the correct prediction (true: Aush, predicted: Aush):



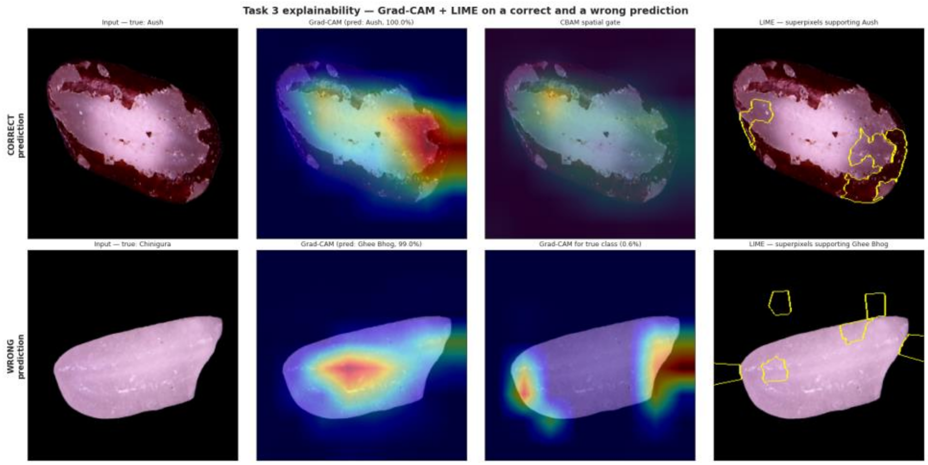
LIME explanation output for the required wrong prediction (true: Chinigura, predicted: Ghee Bhog), generated from the notebook:



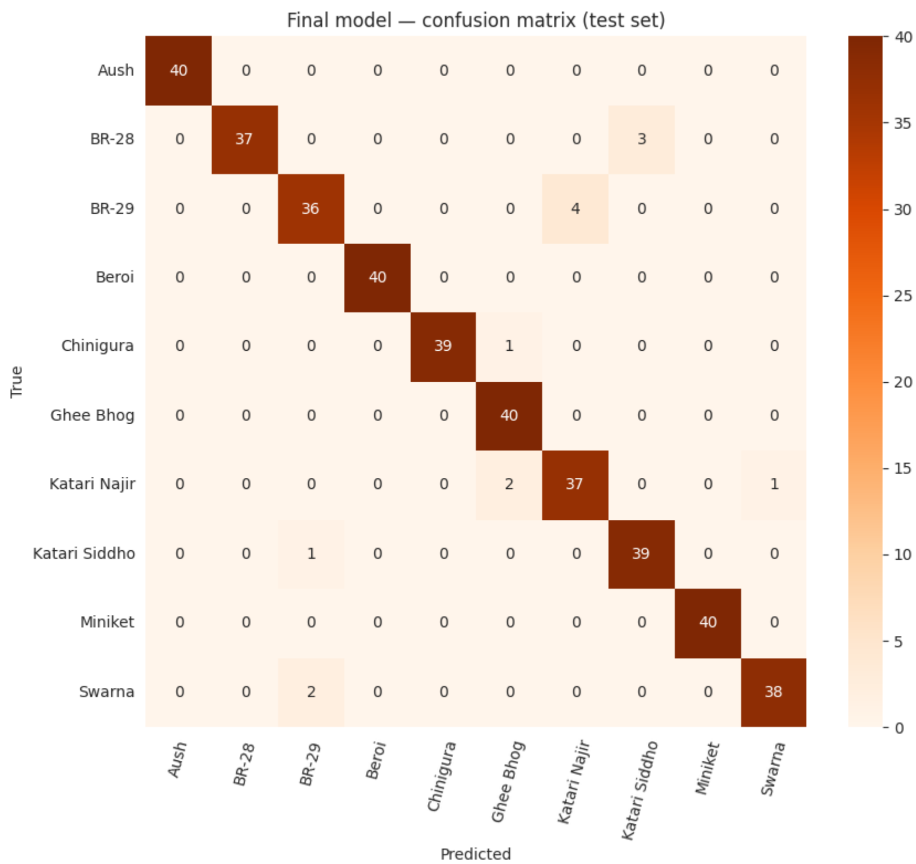
For the correct prediction, Grad-CAM concentrates on the grain body rather than the background, indicating that the decision is driven by rice texture itself; Grad-CAM and LIME show weak-to-moderate agreement (IoU = 0.174 between the LIME superpixel mask and the top-30% Grad-CAM region), which is meaningful corroboration since LIME does not use the model's gradients at all.

For the wrong prediction, the model assigned only 0.6% probability to the true class Chinigura, so this is a confident error rather than a borderline one. Comparing the Grad-CAM map for the predicted class (Ghee Bhog) with the map for the true class (Chinigura) shows the two maps sit in overlapping but not identical regions of the grain, suggesting the model located a plausible region of evidence but misread the fine-grained texture rather than looking at the wrong part of the image entirely. Grad-CAM and LIME again show weak agreement on this image (IoU = 0.080), consistent with the explanation being less stable on a misclassified, harder example.

Combined Grad-CAM and LIME summary panel - correct prediction (top row) versus wrong prediction (bottom row), generated from the notebook:



Confusion matrix on the test set, used for per-class error analysis, generated from the notebook:



The dominant confusion pair is BR-29 being misread as Katari Najir (4 test images), consistent with the fine-grained texture similarity these two varieties share under microscopic imaging. Overall, Grad-CAM and LIME agree that the model's decisions are grounded in the rice grain region rather than background artefacts, and the error analysis indicates the model's mistakes are concentrated among visually similar variety pairs rather than being random.

18. Conclusion

Across all four stages of this project, a complete pipeline was built for rice variety classification from microscopic images. The exploratory data analysis (Task 1) confirmed a clean, perfectly balanced dataset with no exploitable colour or shortcut features, motivating an attention-based approach. Task 2 established four CNN baselines and a ResNet50+CBAM attention model evaluated under a leakage-free protocol, while the supplementary GNN study showed that graph-based models (GCN, GATv2) built on rich node features can match or slightly exceed the strongest CNN baseline while using far fewer parameters. Building on these findings, the Task 3 experiments demonstrate that combining CNN optimization, CBAM attention, rigorous ablation analysis, statistical validation, and explainability creates a reliable deep learning framework for rice variety classification. The final model is not only accurate but also interpretable, allowing better understanding of the visual features used for prediction. Overall, the project shows a clear improvement path from an unbiased understanding of the data, through baseline modelling and attention-based refinement, to a final model that is both statistically validated and explainable resulting in a system well-suited for practical rice variety identification tasks.